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Large Sample Theory | UF Spring 2023

The notes in this PDF are based on the course "Large Sample Theory" taught by Dr. Datta at UF in the Spring semester of 2023. The course, and these subsequent notes, utilize a plethora of text and contains complete exerts from a multitude of boots that are ***not*** referenced in these notes. All information can be found easily online. The list of text can be found in the course syllabus.
A word of caution, these are my personal notes and not a textbook or complete text. These are under the edit of no professor and should be considered as trustworthy as a blog post. It should be noted that if you do not confirm the equations/proofs yourself (which you should for practice) then the notes cannot be used effectively. As Dr. Datta does not provide lecture notes, this is my best rendition.

1 Probability Spaces

A probability space is represented as $(\Omega, \mathcal{F}, P)$.

Ω: is defined as the sample space consisting of elements ω which are considered outcomes of a random experiment.

e.g. Flipping a coin once gives $\Omega = \{H, T\}$ where $\omega_1 = H$, and $\omega_2 = T$.

$\mathcal{F}$: is a σ-algebra, where $\mathcal{F} \subset \mathcal{P}(\Omega)$ (power set). $\mathcal{F}$ is closed under countable unions and compliments (see *ii* and *iii*). Given some $A \in \mathcal{F}$, where A is a Borel Set, some properties of $\mathcal{F}$ include:

(i) $\Omega \in \mathcal{F}$

(ii) $A \in \mathcal{F} \iff A^c \in \mathcal{F}$

(iii) $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

P: is a mapping: $P : \mathcal{F} \to [0, 1]$. A probability mapping follows a measure, μ. This measure in discrete cases is called the <u>Count Measure</u>. In continuous cases it is often the <u>Lebesgue Measure</u>; $\lambda\{[a, b]\} = b - a$.

1.1 Probability Mapping

Some properties of the mapping $P : \mathcal{F} \to [0, 1]$ are:

(i) Let $A \in \mathcal{F}$, then $0 \leq P(A) \leq 1, \forall A \in \mathcal{F}$.

(ii) $P(\Omega) = 1$

(iii) Suppose we have a countable list of events $A_i \in \mathcal{F}, i = 1, 2, \dots$, and when $i \neq j$ $A_i \cap A_j = \emptyset$. Then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

(iv) *Derivation of* $P(\emptyset) = 0$*:* By all previous entries,

$$P(\emptyset \cup \Omega) \underset{\text{by iii}}{=} P(\emptyset) + P(\Omega) \leq 1 \underset{\text{by i and ii}}{\Longrightarrow} 0 \leq P(\emptyset) \leq 1 - P(\Omega) = 0 \Rightarrow P(\emptyset) = 0$$

(v) *Derivation of* $A \subset B \Rightarrow P(A) \leq P(B)$:

$$\text{if } A \subset B, \text{ then } A \cup B \backslash A = B \underset{\text{by iii}}{\Rightarrow} P(A) + P(A \backslash B) = P(B) \underset{\text{by i}}{\Rightarrow} P(A) \leq P(B)$$

where $B \backslash A$ is the set of B excluding all elements from A.

(vi) <u>Subadditivity</u>: Suppose $A_1, A_2, \cdots \in \mathcal{F}$ and $A_i \cap A_j \neq \emptyset$ for at least one $i \neq j$, then

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} A_i\right) &= P(A_1) + P(A_2) - P(A_1 \cap A_2) \\ &+ P(A_3) - P(A_3 \cap P(A_1 \cup A_2)) \\ &+ P(A_4) - P(A_4 \cap P(A_1 \cup A_2 \cup A_3)) + \cdots \leq \sum_{i=1}^{\infty} P(A_i) \end{aligned}$$

1.2 Limits, Supremum, and Infimum

A set is considered to be a <u>monotonically increasing sequence of events</u> if

$$A_1 \subset A_2 \subset \cdots \rightarrow \lim_{n \rightarrow \infty} A_n = \bigcup_{i=1}^{\infty} A_n$$

A set is considered to be a <u>monotonically decreasing sequence of events</u> if

$$B_1 \supset B_2 \supset \cdots \rightarrow \lim_{n \rightarrow \infty} B_n = \bigcap_{i=1}^{\infty} B_n$$

The <u>Limit Supremum</u> or <u>Limsup</u> of a set S, which is a subset of a partial ordered set P, is defined as the smallest element in P that is greater than or equal to all elements in S. Formally this is written as

$$\limsup_{n \rightarrow \infty} A_n = \overline{\lim}_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \underbrace{\left[\bigcup_{k=n}^{\infty} A_k\right]}_{B_n}$$

where B_n is monotonic decreasing. It is also the case that

$$P\left(\overline{\lim}_{n \rightarrow \infty} A_n\right) = \overline{\lim}_{n \rightarrow \infty} P\left(\bigcup_{k=n}^{\infty} A_k\right)$$

The <u>Limit Infimum</u> or <u>Liminf</u> of a set S, which is a subset of a partial ordered set P, is defined as the largest element in P that is less than or equal to all elements in S.
Formally this is written as

$$\liminf_{n \rightarrow \infty} A_n = \underline{\lim}_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \underbrace{\left[\bigcap_{k=n}^{\infty} A_k\right]}_{B_n}$$

where B_n is monotonic increasing. It is also the case that

$$P\left(\underline{\lim}_{n \rightarrow \infty} A_n\right) = \underline{\lim}_{n \rightarrow \infty} P\left(\bigcap_{k=n}^{\infty} A_k\right)$$

2 Random Variables and Distribution

Let's begin with an exmaple, say we have the random variable X. X maps a probability space to $\mathbb{R}$.

$$X : (\Omega, \mathcal{F}, P) \to \mathbb{R}$$

ex. Let's say you toss a coin 10 times, in this case $\omega = \{\text{all possible results from 10 tosses}\}$. From here we can define X as some mapping of the set Ω to $\mathbb{R}$ in value alone. Let's say

X is the number of Heads that occur in the 10 tosses

Now we have a set of $B = \{1, 2, \ldots, 10\}$, which is the set of values $X(\omega)$ may take. In math terms

$$B = \{X(\omega) : \omega \subset \Omega\}$$

B is considered a Borel-Sigma algebra, and thus we have the first two portions of the mapping (Ω and $\mathcal{F}$). Next, the probability induced by X follows by further definition of the events. Let's say the coin is fair, now we may define a <u>Probability Distribution</u> of X.

2.1 Probability Distributions

If X maps a probability space (i.e. $X : (\Omega, \mathcal{F}, P) \to [0, 1]$), we would call $X = P_X$ a Probability Distribution.

$$P_X(B) \overset{\text{def.}}{=} P(X^{-1}(B)) = \{P(\omega);\ \text{where } X(\omega) \in B\} \text{ sometimes written as } P_X$$

We say P_X has a <u>μ-density</u> p_X if

$$p_X = \frac{d}{d\mu} P_X(B) \iff P_X(B) = \int_B p_X d\mu = \int_B I_B(t) p_X(t) d\mu(t)$$

where $d\mu$ is the <u>density measure</u>. For continuous distributions $d\mu = d\lambda$, the Lebesgue Measure, and for discrete distributions $d\mu$ is the Count Measure, as mentioned previously. We call μ a <u>σ-finite dominating measure</u> if

$$\exists A_n \ni \mu(A_n) < \infty\ \forall n \text{ and } \Omega = \bigcup_{n=1}^{\infty} A_n$$

Example: Show $\mu(\mathbb{R}) = \infty$ (show the measure of $\mathbb{R}$ is infinite).
Let $A_n = (-n, n)$. Note that $\mu(A_n) < \infty\ \forall n$. However, as $n \to \infty$, $A_n \to (-\infty, \infty) \equiv \mathbb{R}$ and $\mu(-\infty, \infty) = \lambda(-\infty, \infty) = \infty$

2.2 Distribution Functions

The distribution function can be defined as

$$F_X(t) = P_X(-\infty, t] = P\{X^{-1}(-\infty, t]\} = P\{\omega;\ X(\omega) \leq t\}$$

Properties of $F_X(\cdot)$ include:

(i) $0 \leq F_X(\cdot) \leq 1$

(ii) $F_X(\cdot)$ is monotonic increasing.

(iii) $F_X(-\infty) = 0$, $F_X(\infty) = 1$

(iv) $F_X(\cdot)$ is right-continuous and the left-limit exists at every point.

2.3 Expectation

The expectation can be defined as

$$E(X) = \int_{\Omega} X dP = \int X(\omega) dP(\omega),\ \omega \in \Omega \tag{1}$$

Expectation can also be defined in another way.

$$E(X) = \int t dP_X(t), \quad t \in X(\omega) \tag{2}$$

Example of (1) & (2); 2 tosses of a fair coin. Let $X =$ the number of heads.
(1): $\Omega = \{HH, HT, TH, TT\}$, $P_\omega = \{1/4, 1/4, 1/4, 1/4\}$.

$$E(X) = \int X(\omega) dP(\omega) = 2 * \frac{1}{4} + 1 * \frac{1}{4} + 1 * \frac{1}{4} + 0 * \frac{1}{4} = 1$$

(2): $\{X(\omega)\} = \{0, 1, 2\}$, $P_X(t) = \{1/4, 2/4, 1/4\}$.

$$E(X) = \int t dP_X(t) = 0 * \frac{1}{4} + 1 * \frac{1}{2} + 2 * \frac{1}{4} = 1$$

Expectation can be defined using the Distribution Function as well, albeit given the function satisfies the condition of the Continuous Mapping Theorem (which requires $g(\cdot)$ to be bounded and Lipshitz (continuous). See the following:

$$E(g(X)) = \int \underbrace{g(t)}_{\substack{\text{where} \\ \text{the jumps} \\ \text{happen}}} \underbrace{dF_X(t)}_{\substack{\text{distance} \\ \text{of jumps} \\ \text{in the} \\ \text{distribution} \\ \text{function}}}$$

2.4 Moments & Characteristic functions

The Kth moment of X is defined as

$$\mu_X^k = \int X^k dF(X)$$

A <u>moment generating function</u> (M.G.F.) is

$$E(e^{tX})$$

Recall the <u>characteristic function</u>

$$\phi_X(t) = E(e^{itx}) = E(\cos(tX)) + iE(\sin(tX))$$

For any distribution function the following are true

(i) The characteristic function exists.

(ii) $|\phi_X(t)| \leq E|e^{itX}| = 1$

(iii) $\phi_X(t)$ is continuous and bounded

(iv) if $\mu_k < \infty$, then $\phi_X(t)$ is k-times differentiable and $\phi_X(0)^{(k)} = i^k E(X^k)$

Moment generating functions are not unique, however characteristic functions are unique to distributions.

3 Convergence

Consider a sequence of random variables and a random variable on the following probability spaces:

$$X_n(\Omega_n, \mathcal{F}_n, P_n) \to \mathbb{R} \qquad X(\Omega, \mathcal{F}, P) \to \mathbb{R}.$$

In order of strength of convergence:

(i) $X_n \xrightarrow{C} X$; <u>X_n converges to X completely</u> if

$$\sum P\{|X_n - X| > \epsilon\} < \infty.$$

(ii) $X_n \xrightarrow{a.s.} X$; <u>X_n converges to X almost surely</u> if

$$(\Omega_n, \mathcal{F}_n P_n) = (\Omega, \mathcal{F}, P)\ \forall n$$

and

$$P\{\omega : X_n(\omega) \to X \text{ as } n \to \infty\} = 1$$

(iii) $X_n \xrightarrow{L_r} X$; <u>X_n converges to X in L_r</u> for

$$r \geq 1 \text{ if } E|X_n - X|^r \to 0 \text{ as } n \to \infty$$

(iv) $X_n \xrightarrow{P} X$; <u>X_n converges to X in probability</u> if

$$\text{given any } \epsilon > 0, \text{ and } (\Omega_n, \mathcal{F}_n, P_n) = (\Omega, \mathcal{F}, P)$$

$$\text{then, } P\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\} \to 0, \text{ as } n \to \infty.$$

(v) $X_n \xrightarrow{d} X$; <u>X_n converges to X in distribution</u> iff

$$F_{X_n}(t) \to F_X(t),\ \forall t \text{ (only when } F_X(t) \text{ is continuous).}$$

The <u>Modes of Convergence</u> are represented by a tree:

$$c \implies a.s \iff p \implies d \underset{\text{if constant}}{\implies} p$$

and

$$L_r \underset{r>s}{\implies} L_s \implies p$$

3.1 Useful theorems

Before all the proofs of modes are done, let us introduce some tools to handle the proofs.

3.1.1 Convergence in L_r implies convergence in L_s when $r > s$.

Use Jensen's inequality. Recall absolute value is a convex function.

3.1.2 Markov's Inequality

$$P(|X| \geq a) \leq \frac{E(|X|)}{a}$$

pf:

$$E(X) = \int_\Omega X dP = \int\limits_{\{\omega;\, X(\omega)<a\}} X dP \; + \int\limits_{\{\omega;\, X(\omega)\geq a\}} X dP \; \geq \int\limits_{\{\omega;\, X(\omega)\geq a\}} X dP \; \geq \; a \int\limits_{\{\omega;\, X(\omega)\geq a\}} dP \; = aP(X \geq a)$$

3.1.3 Triangle Inequality

$$P(Y \leq a) \leq P(X \leq a + \epsilon) + P(|X - Y| > \epsilon)$$

3.1.4 Borel Cantelli Lemma

$$\sum_{n=1}^{\infty} P(A_n) < \infty \implies P\left(\limsup_{n\to\infty} A_n\right) = 0$$

3.1.5 Slutsky's Lemma

Let $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{P} C$, where C is non-random. Then

(i) $X_n \pm Y_n \xrightarrow{d} X \pm C$

(ii) $X_n \cdot Y_n \xrightarrow{d} X \cdot C$

(iii) $\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{C}$ where $Y_n, C \neq 0 \; \forall n$

3.2 Proofs of Convergence

3.2.1 Almost sure convergence implies convergence in probability

Consider $\limsup\{|X_n - X| > \epsilon\}$, and let $\{|X_n - X| > \epsilon\} = A_n$. Then we have that,

$$\limsup A_n = \bigcap_{n=1}^{\infty} \left[\bigcup_{k=n}^{\infty} A_k\right] = \bigcap_{n=1}^{\infty} B_n$$

where B_n is monotonically decreasing. However, we note that

$$A_n \subset B_n \implies P(A_n) \leq P(B_n)$$

From the assumption we have that $X_n \xrightarrow{a.s.} X$ which means

$$P\{\omega : X_n(\omega) - X \to 0 \text{ as } n \to \infty\} = 1 \iff P\{\omega : |X_n(\omega) - X| > 0 \text{ as } n \to \infty\} = 0$$

Therefore, $P(B_n) = P\left(\bigcup_{n=k}^{\infty}\{|X_k - X| > \epsilon\}\right) \to 0$ as $n \to \infty$. Therefore,

$$P(A_n) = P(|X_n - X| > \epsilon) \to 0 \text{ as } n \to \infty$$

3.2.2 Convergence in probability implies convergence in distribution

Consider a sequence of r.v.s X_n and a r.v. X with CDFs F_{X_n} and F_X respectively. Consider

$$P(X_n \leq t) \leq P(X \leq t + \epsilon) + P(|X_n - X| > \epsilon)$$

as well as

$$P(X \leq t - \epsilon) \leq P(X_n \leq t) + P(|X_n - X| > \epsilon)$$

If we let $n \to \infty$, then we have that $P(|X_n - X| > \epsilon) \to 0$ by assumption, the inequality follows

$$P(X \leq t - \epsilon) \leq P(X_n \leq t) \leq P(X \leq t + \epsilon)$$

and thus when ϵ approaches 0,

$$F_{X_n} \to F_X \text{ as } n \to \infty.$$

3.2.3 Convergence in L_r implies convergence in probability

Given that $E|X_n - X|^r \to 0$ as $n \to \infty$ we know that $E|X_n - X| \to 0$ as $n \to \infty$. By Markov's inequality for any $\epsilon > 0$ we have that as $n \to \infty$

$$P(|X_n - X| \geq \epsilon) \leq \frac{E|X_n - X|}{\epsilon} \to 0 \implies P(|X_n - X| \geq \epsilon) \to 0$$

and we're done.

3.2.4 Convergence in distribution to a constant implies convergence in probability

Let's assume we have some sequence of random variables X_n and some constant c such that $F_{X_n} \to F_c$ as $n \to \infty$, where $F_c(t) = 0$ if $t < c$, and $F_c(t) = 1$ if $t > c$. Given any $\epsilon > 0$ it can be seen that

$$\begin{aligned} P(|X_n - c| > \epsilon) &= P(X_n - c < -\epsilon) + P(X_n - c > \epsilon) \\ &= F_{X_n}(c - \epsilon) + 1 - F_{X_n}(c + \epsilon) \\ &\xrightarrow{n \to \infty} 0 + 1 - 1 = 0. \end{aligned}$$

3.3 Counter Examples

3.3.1 Convergence in probability does not imply complete convergence

$X_n \xrightarrow{p} X \implies X_n \xrightarrow{c} X$

Consider the sample space $[0, 1]$ with probability measure $P = \lambda$ (the Lebesgue measure). Let

$$X_n(\omega) = 1_{[0, \frac{1}{n})} = \begin{cases} 1 & \omega \in [0, \frac{1}{n}) \\ 0 & \text{o.w.} \end{cases}$$

For some $\epsilon \in (0, 1)$, it is the case that

$$P\{|X_n(\omega)| > \epsilon\} = \frac{1}{n} \xrightarrow{p} 0 \text{ as } n \to \infty.$$

However

$$\sum_{i=1}^{n} P\{|X_n(\omega)| > \epsilon\} = \sum_{i=1}^{n} \frac{1}{n} \text{ diverges as } n \to \infty$$

and we are done.

3.3.2 Convergence in probability does not imply convergence almost surely

$X_n \xrightarrow{p} X \implies X_n \xrightarrow{a.s.} X$

Consider the sample space $(0, 1]$ with probability measure $P = \lambda$ (the Lebesgue measure). Let

$$\begin{aligned}
X_1(\omega) &= 1_{(0,1]}(\omega) \\
X_2(\omega) &= 1_{(0,\frac{1}{2}]}(\omega) \\
X_3(\omega) &= 1_{(\frac{1}{2},1]}(\omega) \\
X_4(\omega) &= 1_{(0,\frac{1}{4}]}(\omega) \\
X_5(\omega) &= 1_{(\frac{1}{4},\frac{1}{2}]}(\omega) \\
&\vdots
\end{aligned}$$

Then X_n does not converge almost surely because for any given $\omega \in (0, 1]$ and $N \in \mathbf{N}$, there exists some $n, m > N$ such that $X_n(\omega) = 1$ and $X_m(\omega) = 0$. However, for any $\epsilon > 0$ it is the case that $P\{|X_n(\omega)| > \epsilon\} \leq \frac{2}{n+1} \to 0$ as $n \to \infty$.

3.3.3 Convergence in distribution does not imply convergence in probability

$X_n \xrightarrow{d} X \implies X_n \xrightarrow{p} X$

Consider $Z = N(0, 1)$ and $-Z$. It is the case that $-Z \xrightarrow{d} Z = N(0, 1)$. However, given some $\epsilon > 0$,

$$P(|-Z - Z| > \epsilon) = P\{2Z > \epsilon\} \not\to 0.$$

Hence, $-Z \overset{p}{\not\to} Z$.

4 O-notation

"Little o": If $X_n = o_p(1)$, then X_n converges in probability to 0.
"Big O": If $X_n = O_p(1)$ then there exists an interval, $(-m_\epsilon, m_\epsilon)$, in which the distribution function, F_{X_n}, is contained. This is sometimes referred to as X_n being "tight".

$$X_n = o_p(1) := X_n \xrightarrow{P} 0 \text{ and } X_n = o_p(C_n) := \frac{X_n}{C_n} \xrightarrow{P} 0$$

$$X_n = O_p(1) := \text{ for any } \epsilon > 0,\ \exists m_\epsilon < \infty \text{ s.t. } P\{|X_n| \leq m_\epsilon\} > 1 - \epsilon\ \forall n \geq 1$$

note: a convergent sequence is always bounded, however a bounded sequence is not always convergent.

$$X_n \to X \implies X_n = O_p(1)$$
$$X_n = O_p(1) \not\Rightarrow X_n \to X$$

<u>Counter Example</u>: Let $X_n = (-1)^n$. $-1 \leq X_n \leq 1$, so $m_\epsilon = 1$. However, as $n \to \infty$, X_n does not converge.
We can use this notation as a means to show the results of linear transformations of arbitrary sequences based on their convergence. For example,

$$\text{If } X_n = O_p(1),\ Y_n = o_p(1), \text{ then } X_n \pm Y_n = O_p(1)$$
$$\text{If } X_n = O_p(1),\ Y_n = O_p(1), \text{ then } X_n \pm Y_n = O_p(1)$$

If $C_n,\ D_n$ are sequences in $\mathbb{R}$

$$X_n = O_p(C_n) \iff \frac{X_n}{C_n} = O_p(1)$$
$$X_n = o_p(C_n) \iff \frac{X_n}{C_n} = o_p(1)$$
$$O_p(C_n) + O_p(D_n) = O_p(C_n + D_n)$$

provided $C_n \neq 0$.

5 Introduction Asymptotics

5.1 Law of Large Numbers

The law of large numbers is as follows. Given $x_1, x_2, \ldots$ i.i.d., $E(x_i) = \mu \in \mathbb{R}$, let $\bar{x} = \bar{x}_n = \frac{x_1 + \cdots + x_n}{n} = \frac{S_n}{n}$. Then,

$$\bar{x_n} \xrightarrow{a.s.} \mu \qquad \text{Strong LLN}$$
$$\bar{x_n} \xrightarrow{p} \mu \qquad \text{Weak LLN}$$

5.2 Central Limit Theorem (CLT)

Given x_i's i.i.d. and $E(x) = \mu$, $var(x) = \sigma^2 < \infty$, then,

$$Z_n = \frac{\bar{x}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \to Z \sim N(0, 1)$$

the multivariate version of the CLT can be seen online, but is quite simple.
`https://en.wikipedia.org/wiki/Central_limit_theorem#:~:text=Multidimensional%20CLT`

5.2.1 Quick proof of CLT

Without loss of generality let $\mu = 0$, $\sigma = 1$. Then all we have to show is that

$$\sqrt{n}\bar{x}_n \to Z \sim N(0, 1)$$

Let's look at the characteristic function of Z_n.

$$\phi_{Z_n}(t) = E\{\exp(itZ_n)\} = E\{\exp(it(\sqrt{n}\bar{x_n})\} = \prod_{i=1}^{n} \phi_{X_i}(t/\sqrt{n}) = \left\{\phi_{X_1}(t/\sqrt{n})\right\}^n$$

(the last equality is from i.i.d.) From here all we need to show is

$$\left\{\phi_{X_1}(t/\sqrt{n})\right\}^n \to \{\phi_{X_1}(0)\}^n$$

Using the taylor expansion on the LHS,

$$\phi_{X_1}(t/\sqrt{n})\Big|_0 = 1 - t^2/2n + o(1/n)$$

So we have that

$$\phi_{Z_n}(t) = (1 - t^2/2n)^n \xrightarrow{n\to\infty} \exp(-t^2/2)$$

which is the characteristic function for standard normal.

5.2.2 Asymptotic notation

Here is a bit of notation that will be used later on

$$X_n \sim AN(\mu_n, \sigma_n^2)$$

where X_n is a sequence of random variables and AN stands for asymptotically normal. It means

$$Z_n := \frac{X_n - \mu_n}{\sigma_n} \xrightarrow{d} N(0,1)$$

It does NOT mean $E(X_n) = \mu_n$. it does NOT mean $Var(X_n) = \sigma_n$.

Note: the mean cannot depend on n. An example of where this doesn't work is $\bar{X} \sim AN(\mu + \frac{1}{\sqrt{n}}, \frac{\sigma^2}{n})$.
This doesn't work because

$$\frac{X_n - \mu_n}{\sigma_n} = \frac{\bar{X} - \mu + \frac{1}{\sqrt{n}}}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mu + \frac{1}{\sqrt{n}}}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} + \sigma$$

5.3 Taylor Series

Let f be some infinitely differentiable function at a real or complex number x_0. Then $f(x)$ can be written as

$$f(x) = \frac{f^{(0)}(x_0)}{0!}(x - x_0)^0 + \frac{f^{(1)}(x_0)}{1!}(x - x_0)^1 + \frac{f^{(2)}(x_0)}{2!}(x - x_0)^2 + \cdots$$

5.4 Delta Method

Let X_i be IID, $E(X_i) = \mu$, and $V(X_i) = \sigma^2 < \infty$ for $i = 1, \ldots, n$. By CLT it is the case that $\overline{X} \sim AN(\mu, \sigma^2/n)$. Let $g(\cdot) : \mathbb{R} \to \mathbb{R}$, where $g'(\mu) \neq 0$.

Because g is differentiable at μ, we can use the Taylor series to express $g(\overline{X})$ as

$$g(\overline{X}) = g(\mu) + g'(\mu)(\overline{X} - \mu) + o_p(1/\sqrt{n})$$
$$\sqrt{n}\{g(\overline{X}) - g(\mu)\} = \sqrt{n}(\overline{X} - \mu)g'(\mu) + o_p(1)$$

CLT tells us that

$$\sqrt{n}(\overline{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

and we know that $g'(\mu)$ is some non-zero constant. So, by Slutsky's lemma,

$$\sqrt{n}(\overline{X} - \mu)g'(\mu) \xrightarrow{d} g'(\mu)N(0, \sigma^2),$$

and by equality

$$\sqrt{n}\{g(\overline{X}) - g(\mu)\} + o_p(1) \xrightarrow{d} N(0, g'(\mu)^2\sigma^2).$$

(note that $o_p(1)$ can be ignored as $o_p(1) \to 0$ quite fast). The multivariate version of delta method is

$$\sqrt{n}\{g(\overline{X}) - g(\mu)\} \xrightarrow{d} N(0, \nabla g(\mu)^T \Sigma \nabla g(\mu)).$$

6 Influence functions: the backbone of asymptotic statistics

It's important to get a grasp on the way that CLT and Delta method function in regard to Influence Functions. Let's assume we are interested in estimating a parameter θ for an iid set of data. A *centered* influence function, ϕ, takes the form of

$$(\hat{\theta} - \theta) = -\frac{1}{n}\sum_{i=1}^{n} \phi^c(x_i) + o_p(n^{-1/2})$$

where c superscript indicates *centered.* In general, CLT and Slutsky's theorem allow us to state that

$$\sqrt{n}(\hat{\theta} - \theta) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \phi^c(x_i) + o_p(1) \xrightarrow{d} N(0, Var(\phi^c(x_1)))$$

we will see this in various situations going forward.
A simple example would be the sample mean for an IID sample with defined variance σ^2.

$$\sqrt{n}(\bar{x} - \mu) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}(x_i - \mu) \xrightarrow{d} N(0, Var(x_1 - \mu)) = N(0, \sigma^2)$$

which is directly stated in the CLT as

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{(x_i - \mu)}{\sigma} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

6.1 Quantiles

Let $0 < p < 1$, the pth quantile, ξ_p, of a CDF is defined as

$$\xi_p = \inf_x \{x : \ F_n(x) \geq p\}$$

or asymptotically equivalent,

$$\xi_p = \sup_x \{x : \ F_n(x) \leq p\}.$$

We can approximate $\hat{p}$ with p by using the Taylor expansion on the function that defines the empirical probability,

$$\hat{p} = F(\hat{\xi}_p) = \frac{1}{n}\sum_{i=1}^{n} I(x_i \leq \hat{\xi}_p), \text{ and } p = F(\xi_p) = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} I(x_i \leq \xi_p)$$

and

$$F(\hat{\xi}_p) = F(\xi_p) - (\hat{\xi}_p - \xi_p)F'(\xi_p) + o_p(n^{-1/2}) \implies \hat{\xi}_p = \xi_p - \frac{(\hat{F}(\xi_p) - F(\xi_p))}{F'(\xi_p)} + o_p(n^{-1/2})$$

So then,

$$\hat{\xi}_p = \xi_p - \frac{1}{n}\sum_{i=1}^{n} IF(x_i) + o_p(n^{-1/2})$$

where

$$IF(x_i) = \frac{I(x_i \leq \xi_p) - p}{f(\xi_p)}$$

and F has positive density $f(\cdot)$ which is positive on ξ_p.

Example: Sample median, m, for $p = \frac{1}{2}$. We want to estimate $\hat{m}$.

$$\hat{m} = \hat{\xi}_{\frac{1}{2}} = \xi_{\frac{1}{2}} + \frac{1}{n}\sum_{i=1}^{n} \frac{I(x_i \leq m) - \frac{1}{2}}{f(\xi_{\frac{1}{2}})} + o_p(n^{-1/2})$$

The idea is we can 'standardize' $\hat{\xi}_{\frac{1}{2}}$, with the expected value $E(\hat{\xi}_{\frac{1}{2}}) = \xi_{\frac{1}{2}}$ and variance

$$\begin{aligned}
Var(\hat{\xi}_{\frac{1}{2}}) &= Var\left(\frac{I(x_1 \leq m) - \frac{1}{2}}{f(\xi_{\frac{1}{2}})}\right)/n \\
&= Var\left(\frac{I(x_1 \leq m)}{f(\xi_{\frac{1}{2}})}\right)/n \\
&= \frac{1}{nf(\xi_{\frac{1}{2}})^2} Var\left(I(x_1 \leq m)\right) \\
&= \frac{1}{nf(\xi_{\frac{1}{2}})^2}(1/2)(1 - 1/2) \\
&= \frac{1}{4nf(\xi_{\frac{1}{2}})^2}.
\end{aligned}$$

Finally CLT gives us that

$$\sqrt{n}(\hat{\xi}_{\frac{1}{2}} - \xi_{\frac{1}{2}}) \xrightarrow{d} N\left(0, \frac{1}{4f(\xi_{\frac{1}{2}})^2}\right).$$

which is equivalent to saying

$$\hat{\xi}_{\frac{1}{2}} \sim AN\left(\xi_{\frac{1}{2}}, \frac{1}{4nf(\xi_{\frac{1}{2}})^2}\right).$$

Therefore, the asymptotic variance of the median quantile is $Var(\hat{\xi}_{\frac{1}{2}}) = \frac{1}{4nf(\xi_{\frac{1}{2}})^2}$

7 Maximum Likelihood

Let $X_1, \ldots, X_n \overset{iid}{\sim} F(\theta)$, with $\theta \in \Theta \subset \mathbb{R}^p$. Let $p = 1$.
The likelihood function is treated as a function of θ.

$$L_n(\theta) = \prod_{i=1}^{n} P_\theta(x_i)$$

The log-likelihood function is treated as a function of θ.

$$l_n(\theta) = \sum_{i=1}^{n} \log P_\theta(x_i)$$

The <u>maximum likelihood estimator</u> or <u>MLE</u> of θ is

$$\hat{\theta}_{\text{MLE}} = \underset{\theta}{\text{argmax}}(l_n(\theta))$$

Let Θ be closed and bounded (compact), and let $P_\theta(x)$ be continuous for all $\theta \in \Theta$. The following is the score method of calculating the MLE. If P_θ is differentiable on Θ, then one can try solving the estimating equation

$$S_n(\theta) = 0,$$

where

$$S_n(\theta) = \sum_{i=1}^{n} s_i(\theta), \quad s_i(\theta) = \frac{\partial}{\partial\theta} \log P_\theta(x_i)$$

Say $\hat{\theta}_n$ is a solution, such that $S_n(\hat{\theta}_n) = 0$.

We know $\hat{\theta}_n$ is a minimizor or maximizor depending on $S_n'(\hat{\theta}_n)$ relative to 0. If $S_n'(\hat{\theta}_n) < 0$ it will maximize, otherwise it will minimize. Moreover, if $S_n'(\hat{\theta}_n) < 0$, the only thing we can tell from $S_n(\hat{\theta}_n) = 0$ is that $\hat{\theta}_n$ is a <u>local maximum</u>. In order to make sure $\hat{\theta}_n$ is the absolute maximum we have two choices,

1. Make sure S_n has a unique root (one root only). - this can be difficult
2. Make sure $\hat{\theta}_n \xrightarrow{p} 0$. - this is usually much easier

If we are in the case where we have multiple estimators that solve the score equation, and another estimator that is considered consistent, we would always choose the estimator which was closest to the consistent estimator.

7.1 Proving Consistency for MLE

Let's assume we;re in the space with $X_1, \ldots, X_n IID \sim P,\ P = P_\theta$.
Given that $\hat{\theta}_n = \underset{\theta}{\text{argmax}} \sum \log P_\theta(x_i) = \underset{\theta}{\text{argmax}}\ l(\theta)$, as well as $\theta_0 \in \Theta$ is the true value of θ. We define

$$\begin{aligned}
\Delta_n(\theta) &= \frac{1}{n}\{l(\theta) - l(\theta_0)\} \\
&= \frac{1}{n}\left\{\sum \log P_\theta(x_i) - \sum \log P_{\theta_0}(x_i)\right\} \\
&= \frac{1}{n}\sum \log\left\{\frac{P_\theta(x_i)}{P_{\theta_0}(x_i)}\right\}
\end{aligned}$$

such that

$$\hat{\theta}_{\text{MLE}} = \underset{\theta}{\text{argmax}} \Delta_n(\theta).$$

Looking at the limit of $\Delta_n(\theta) \to \Delta(\theta) = E_\theta\left[\log\left\{\frac{P_\theta(x_1)}{P_{\theta_0}(x_1)}\right\}\right]$, we can use Jensen's inequality due to the concavity of the logarithm function.

$$\begin{aligned}
\Delta(\theta) &\leq \log\left[E_\theta\left\{\frac{P_\theta(x_1)}{P_{\theta_0}(x_1)}\right\}\right] \\
&= \log\left[\int_\Theta \frac{P_\theta(x_1)}{P_{\theta_0}(x_1)} P_{\theta_0}(x_1) dx_1\right] \\
&= \log\left[\int_\Theta P_\theta dx_1\right] \\
&= \log(1) \\
&= 0
\end{aligned}$$

and so we have that $\Delta_n(\theta) \to \Delta(\theta) \leq 0$ and we know that $\Delta(\theta_0) = 0$. We then have that $\Delta(\theta_0)$ must equal $\Delta(\theta)$ and $\Delta_n(\theta) \to \Delta(\theta_0)$. The only thing preventing us from claiming that $\hat{\theta}_{\text{MLE}} \to \theta_0$ is that we don't know if the function $\Delta_n(\theta) \to \Delta(\theta_0)$ <u>converges uniformly</u>. If we assume that it does then we are done.

7.2 MLE Asymptotic distribution

Assume we have $\hat{\theta}_n \to \theta_0$ and $\hat{\theta}_n$ is a consistent estimator where $S_n(\hat{\theta}_n) = 0$. Then,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0))$$

where

$$I(\theta_0) = -E_{\theta_0}\left[\frac{\partial^2}{\partial\theta^2}\log(P_\theta(x_1))\bigg|_{\theta=\theta_0}\right] = E_{\theta_0}\left(\left[\frac{\partial}{\partial\theta}\log(P_\theta(x_1))\bigg|_{\theta=\theta_0}\right]^2\right)$$

Recall that $S_n(\theta) = \sum_{i=1}^n s_i(\theta) = \sum_{i=1}^n \frac{\partial}{\partial\theta}\log(P_\theta(x_i))$. So we can write $I(\theta)$ as

$$I(\theta_0) = -E_{\theta_0}\left[\frac{\partial}{\partial\theta}s_1(\theta)\bigg|_{\theta=\theta_0}\right] = E_{\theta_0}\left(s_1(\theta_0)^2\right)$$

We may also want to show un-biased-ness for the MLE estimator. We show this by using the definition of expectation.

$$\begin{aligned}
E(S_n(\theta)) &= \sum_{i=1}^{n} E_{\theta_0}(s_i(\theta)) \\
&= \sum_{i=1}^{n} E_{\theta_0}\left(\frac{\partial}{\partial\theta}\log(P_\theta(x_i))\right) \\
&= \sum_{i=1}^{n} \int_\Theta \left.\frac{P'_\theta(x_i)}{P_\theta(x_i)}\right|_{\theta=\theta_0} P_{\theta_0}(x_i)dx_i \\
&= \sum_{i=1}^{n} \frac{\partial}{\partial\theta}\int_\Theta P_{\theta_0}(x_i)dx_i \\
&= \sum_{i=1}^{n} 0 \\
&= 0.
\end{aligned}$$

Now for the proof of the asymptotic statement above. We are given $X_1, \ldots, X_n$ IID $\sim$ P_{θ_0}, $\hat{\theta}_n \to \theta_0$, and $\hat{\theta}_n$ is a consistent estimator such that $S_n(\hat{\theta}_n) = 0$. Consider

$$\begin{aligned}
0 &= S_n(\hat{\theta}_n) \\
0 &= S_n(\theta_0) + (\hat{\theta}_n - \theta_0)S'_n(\tilde{\theta}) \qquad (Taylor + MVT) \\
(\hat{\theta}_n - \theta_0) &= -\frac{S_n(\theta_0)}{S'_n(\tilde{\theta})} \\
\sqrt{n}(\hat{\theta}_n - \theta_0) &= -\frac{\frac{1}{\sqrt{n}}S_n(\theta_0)}{\frac{1}{n}S'_n(\tilde{\theta})}
\end{aligned}$$

where $\tilde{\theta}$ is in between θ_0 and $\hat{\theta}_n$. Recall that $S_n(\theta_0) = \sum_{i=1}^{n} s_i(\theta)$ where $s_i(\theta)$ has mean 0 for all $i = 1, \ldots n$ (from independence of $X_i's$). If we let the second moment $E_{\theta_0}\left(s_1(\theta_0)^2\right) = I(\theta_0)$, then by CLT,

$$\frac{1}{\sqrt{n}}(S_n(\theta_0)) = \sqrt{n}\left(\frac{1}{n}\sum s_i(\theta_0)\right) \xrightarrow{d} N(0, I(\theta_0))$$

Recall that consistency of the estimator requires that $\hat{\theta}_n \xrightarrow{u} \theta_0$ and if $\tilde{\theta}$ is in between θ_0 and $\hat{\theta}_n$ it follows that $\tilde{\theta} \xrightarrow{u} \theta_0$, then we have that $S_n(\tilde{\theta}) \to S_n(\theta_0)$ and

$$-\frac{1}{n}S'_n(\tilde{\theta}) \xrightarrow{p} -E_{\theta_0}\left[\left.\frac{\partial^2}{\partial\theta^2}\log(P_\theta(x_1))\right|_{\theta=\theta_0}\right] = I(\theta_0).$$

If we apply slutsky's theorem, we then have that

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\frac{\frac{1}{\sqrt{n}}S_n(\theta_0)}{\frac{1}{n}S'_n(\tilde{\theta})} \xrightarrow{d} \frac{N(0, I(\theta_0))}{I(\theta_0)} = N(0, I^{-1}(\theta_0))$$

that is,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)).$$

7.3 Estimation of the Asymptotic variance of the MLE

The asymptotic variance is a follows

$$A - Var(\hat{\theta}_{\text{MLE}}) = n^{-1} I^{-1}(\theta)$$

The estimator of the asymptotic variance is

$$A - \hat{Var}(\hat{\theta}_{\text{MLE}}) = n^{-1} I^{-1}(\hat{\theta}_{\text{MLE}}) = \frac{1}{n} \hat{B}^{-1} \hat{M} \hat{B}$$

where

$$B = \frac{1}{n} \sum_{i=1}^{n} \frac{\partial^2}{\partial \theta^2} \log P_\theta(x_i) \Big|_{\theta = \hat{\theta}_{\text{MLE}}}$$

$$M = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{\partial}{\partial \theta} \log P_\theta(x_i) \Big|_{\theta = \hat{\theta}_{\text{MLE}}} \right]^2$$

8 Estimating Equations

Ok by this point the course got super confusing and I lost all hope and I needed to get back into it so we start from extreme simplicity. What is an estimating equation?
Consider you have data and an unknown parameter θ. You ultimately want to invent a function, $\psi(\theta)$, such that

- $\psi(\theta)$ is something you can compute from the data for any proposed θ. Think *mean* or *variance.*
- The true parameter θ_0 makes $\psi(\theta_0) = 0$ on average.

So we define the estimating equation as such,

$$\psi(\hat{\theta}) = 0. \tag{3}$$

Here's a simple example with the mean. Let $Y_1, \ldots, Y_n$ be *i.i.d.* with mean μ. A natural estimating equation is going to be a function of μ, the thing we want to compute. Our equation has to satisfy the two requirements above, so we pick

$$\psi(\hat{\mu}) = \sum_{i=1}^{n} (Y_i - \hat{\mu}) = 0.$$

Let's check it works,

1. Yep, easy to compute.
2. If we plugged in the Y values, this would trivially be 0; so that works too.

Great, now we can do what is the main function, to actually solve for what μ is!

$$\begin{aligned}
\psi(\hat{\mu}) = \sum_{i=1}^{n}(Y_i - \hat{\mu}) &= 0 \\
\sum_{i=1}^{n}(Y_i) - \sum_{i=1}^{n}(\hat{\mu}) &= 0 \\
\sum_{i=1}^{n}(Y_i) - n(\hat{\mu}) &= 0 \\
\sum_{i=1}^{n}(Y_i)/n &= \hat{\mu} \\
\bar{Y} &= \hat{\mu}.
\end{aligned}$$

Now, this seems really trivial, why even have estimating equations if we are just solving for the thing we make the equation for? wouldn't it be easier to just calculate it? Let's say there is something really complex we want to estimate and back-solving just isn't that easy. Then we have other methods, like Newtons method, that allow us to solve for solutions of functions that cannot be solved for in a closed form.
Imagine we have $\psi(\theta)$ such that $\psi(\cdot)$ has no closed form solution. Newton's method tells us that,

$$\theta^{(t+1)} = \theta^{(t)} - \frac{U(\theta^{(t)})}{U'(\theta^{(t)})} \tag{4}$$

where if you conveniently add some values of $\sqrt{n}$ and know some neat truths about θ, you can show distributional convergence.
OK, that's the extreme simple case gone, back to the real notation and what the course teaches.

Here we do basically the same thing as we did with the score but in a more general sense. We have the assumptions that $X_1, \ldots, X_n$ IID. We have a general equation where we are solving

$$\sum_{i=1}^{n} \psi(x_i, \theta) = 0$$

Example:

$$\psi(x_i, \theta) = \frac{\partial}{\partial \theta} \log P_\theta(x_i)$$

Can also include covariance such as z_i.

$$\psi(x_i, z_i, \theta)$$

A good way to think about the estimating equation is as any influence function. For example, we want to solve

$$\sum_{i=1}^{n} \psi(x_i, \theta) = 0$$

We know that to make a summation converge in distribution we need it to have mean zero (the estimating equation must be unbiased), in addition we need that the second moment have some sort of estimate that is attainable (the function must be continuous and differentiable) so that we can use Taylor's expansion (Newton's method) to get the form

$$\sqrt{n}(\hat{\theta}-\theta) = \frac{\frac{1}{\sqrt{n}}\sum \psi(x_i,\theta) \xrightarrow{d} N(0,M)}{-\frac{1}{n}\sum \psi'(x_i,\theta) \xrightarrow{p} -B} \xrightarrow{\text{Slutsky's}} N(0,(B^{-1})^t M B^{-1})$$

where

$$M = Var(\psi(x_1,\theta))$$
$$B = E_\theta[\psi'(x_1,\tilde{\theta})\big|_{\tilde{\theta}=\theta}]$$

8.1 Example for estimating equations

Here is an example from the homework:

Consider a simple linear regression model with zero intercept where

$$Y_i = \beta x_i + \epsilon_i$$

where the $x_i's$ are observed covariates and ϵ_i are IID error terms. Let's assume that ϵ_i have a symmetric distribution (note here we have symmetric, this will be important later). Consider estimating the regression parameter β using an estimating equation of the form

$$\sum_{i=1}^{n} \psi(Y_i - \beta x_i)x_i = 0$$

(i) **Show the estimating equation is unbiased for zero under a simple condition on ψ.**

To show the function is unbiased (has zero mean), use the definition of expectation for a continuous function. The expectation is relative to β here because that is what we are estimating. Also, it is assumed that β is continuously distributed and has some distribution function. The distribution function of β is relative to ϵ from the initial equation; it's symmetric.

$$E_\beta(\psi(Y_1-\beta x_1))\gamma_x = E_\beta(\psi(\epsilon_1))\gamma_x = \gamma_x \int_\Theta \psi(\epsilon_1) f_\beta(Y)dY = \gamma_x \int_\Theta \psi(\epsilon_1) f_\beta(Y)dY \stackrel{?}{=} 0$$

where γ_x is the Cesàro summation of the covariates, $\gamma_x = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} x_i$.
If the distribution function is symmetric around the mean, then $f_\beta(Y)$ is *even*. In order for the integral to be equal to 0 when integrating over the entire sample space the integrand must be an *odd function*. Thus, ψ must be an odd function.
This may be complex at the beginning, but think about linear regression! What is the estimating equation for β? It's just,

$$\sum_{i=1}^{n}(Y_i - \beta x_i)x_i = 0,$$

so $\psi(\cdot)$ is just the identify function $f(x) = x$. The identity function, for example $f(y) = x$ is linear, it's a line through the origin, therefore it's odd. Simple enough. If we extent this to be any function of $\psi(\cdot)$, then we demand that function still be odd.

(ii) **Find the Asymptotic distribution of $\hat{\beta}$ including the variance estimation.**

If we assume $\hat{\beta}$ is a consistent estimator such that $\hat{\beta} \xrightarrow{u} \beta$ then we can use Taylor series to establish the equation below.

$$\sqrt{n}(\hat{\beta} - \beta) = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(Y_i - \beta x_i) x_i}{-\frac{1}{n} \sum_{i=1}^{n} \psi'(Y_i - \tilde{\beta} x_i) x_i^2}$$

where $\tilde{\beta}$ is between β and $\hat{\beta}$. Because $\psi(Y_i - \beta x_i)x_i$ is unbiased for 0, by CLT

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(Y_i - \beta x_i) x_i \xrightarrow{d} N(0, \mathbf{C}_x Var_\beta(\psi(\epsilon_1))) = N(0, \mathbf{C}_x E_\beta(\psi(\epsilon_1)^2))$$

where $\mathbf{C}_x = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} x_i^2$. As well we have that

$$\frac{1}{n} \sum_{i=1}^{n} \psi'(Y_i - \tilde{\beta} x_i) x_i^2 \xrightarrow{p} E_\beta(\psi'(\epsilon_1)) \mathbf{C}_x$$

Hence, by Slutsky's theorem we have the asymptotic distribution of $\hat{\beta}$:

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N\left(0, \frac{E_\beta(\psi(\epsilon_1)^2)}{\mathbf{C}_x [E_\beta(\psi'(\epsilon_1))]^2}\right).$$

The estimator for the Asymptotic variance follows as

$$\hat{A} - \hat{Var}(\hat{\beta}) = \frac{\frac{1}{n} \sum_{i=1}^{n} [\psi(Y_i - \hat{\beta} x_i)]^2}{\left[\frac{1}{n} \sum_{i=1}^{n} \psi'(Y_i - \hat{\beta})\right]^2 \sum_{i=1}^{n} x_i^2}$$

9 Statistical Tests

Let's assume we have a sample $X_1, \ldots, X_n$ IID $\sim P_\theta$ where $\theta \in \Theta$. Let's assume we are testing the hypotheses

$$H_0 : \theta = \theta_0 \qquad H_1 : \theta \neq \theta_0$$

1. <u>Likelihood ratio test: LRT</u>
   The likelihood ratio test compares the value of the likelihood functions at the hypothetical value versus the true value. If the ratio is close to one, then the estimator is good. The test is exactly,
   $$-2 \log \Lambda = -2 \log \frac{L_n(\theta_0)}{L_n(\theta)} \sim \chi_p^2(1 - \alpha)$$
   It can be noted that the Leymann-Pearson lemma states this is the most powerful test among all α tests.

2. <u>Score test</u>
   The score test is an extension of the relationship of the influence function to the normal distribution (not we do not actually need to calculate the estimate for the parameter here, nor do we need to calculate the likelihood functions, only their derivatives).

$$\left|\frac{\frac{1}{\sqrt{n}}S_n(\theta_0)}{\sqrt{I(\theta_0)}}\right| > Z_{1-\frac{\alpha}{2}} \iff \frac{S_n^T(\theta_0)I^{-1}(\theta_0)S_n(\theta_0)}{n} > \chi_p^2(1-\alpha)$$

3. <u>Wald test</u>
   The wald test simply compares the the asymptotic distribution of the parameters to a normal distribution.

$$|\sqrt{I(\theta_0)}\sqrt{n}(\theta-\theta_0)| > Z_{1-\frac{\alpha}{2}} \iff n(\theta-\theta_0)^T I(\theta_0)(\theta-\theta_0) > \chi_p^2(1-\alpha)$$

10 U-statistic

The U-statistic section of the course is very straight forward so I will not be writing up a lot and instead will provide some examples from previous home-works

The general setup for the U-statistic and V-statistic is the following: Given an IID sample $\{X_1, \dots X_n\}$ with F distribution, first moment μ and second central moment σ^2, let $h_F(X_{i1}, \dots, X_{im}) : \mathbb{R}^m \to \mathbb{R}$ be symmetric for all $X_{i1}, \dots, X_{im}$, and let $n \geq m$ and m fixed. Let's assume that h is a function that is used to estimate some statistic, such as mean of variance, then the U-statistic and V-statistic are respectively

$$U_m = \frac{1}{\binom{n}{m}} \sum_{1 \leq i_1 < \dots < i_m \leq n} h_F(X_{i1}, \dots, X_{im})$$
$$V_m = \frac{1}{nm} \sum_{i_1=1}^{n} \cdots \sum_{i_m=1}^{n} h_F(X_{i1}, \dots, X_{im})$$

where h is called the *kernel* and is a known function such that,

$$E(h_F(X_1, \dots, X_m)) = \theta.$$

All U statistics are inherently unbiased, hence the name "Unbiased Statistic". It's close cousin, the V statistic is not guaranteed to be unbiased, and is often biased, however V is consistent, so it is still useful as an estimator (i.e. $bias \to 0$ as $n \to \infty$).
A general technique in making h symmetric is to use the following technique and scale it up depending on the size of m.

$$\begin{aligned}
h^*(X_1, X_2) &= X_1^2 - X_1X_2 \qquad \text{non-symmetric for } X_1, X_2 \\
h(X_1, X_2) &= \frac{h^*(X_1, X_2) + h^*(X_2, X_1)}{2} \\
&= \frac{X_1^2 - X_1X_2 + X_2^2 - X_1X_2}{2!} \\
&= \frac{1}{2}(X_1 - X_2)^2 \qquad \text{symmetric for } X_1, X_2.
\end{aligned}$$

In order to get the U statistic in a form that we can utilize what we've learned about influence functions we have to use a Hajek projection of the U statistic onto the set of all statistics that take the form of an influence function (see Van der Vaart). That is,

$$U - \theta \simeq \hat{U} = \sum_{i=1}^{n} E(U - \theta | X_i = x_i) = \frac{m}{n} \sum_{i=1}^{n} h_1^c(x_i)$$

where

$$h_1^c(x_1) = E(h(X_{i1}, \ldots, X_{im}) | X_i = x_1) - \theta = E(h(x_1, X_{i2}, \ldots X_{im})) - \theta.$$

To understand what's going on here it's best to think in face value definitions. We want the expected value of this h function based on some value when one of the random variables in the sample $\{X_1, \ldots, X_n\}$ is fixed at x.

$$E(h(X_{i1}, \ldots, X_{im}) | X_i = x_1) = \begin{cases} E(h(x_1, X_{i2}, \ldots X_{im})) - \theta & \text{if } X_i \in \{X_{i1}, \ldots, X_{im}\} \\ 0 & \text{if } X_i \notin \{X_{i1}, \ldots, X_{im}\} \end{cases}$$

In the first case, if X_i is in the set we are considering for h, because h is symmetric we can just replace the first variable every time and we'll get the same result m many times. As for the second case, if X_i is not in the set $\{X_{i1}, \ldots, X_{im}\}$, then we will have h and θ cancel, since they're the same under the expectation! (recall $E(h_F(X_{i1}, \ldots, X_{im})) = \theta$)

The notation is as follows

$$\begin{aligned} \hat{U} &:= \text{ The projection of U} \\ U - \theta &:= \text{ Asymptotically equivalent to the projection of U} \\ \hat{U} \simeq U - \theta \qquad & \text{and} \qquad \hat{U} + o_p\left(\frac{1}{\sqrt{n}}\right) = U - \theta \end{aligned}$$

As it may be obvious now, the projection looks just like an influence function, thus all the similar asymptotic we're used to follow suit.

$$U - \theta \sim AN\left(0, \frac{m^2}{n}\xi_1^2\right)$$

where

$$Var(h_1^c(x_1)) = \xi_1^2$$

10.1 U statistic example 1

Given an IID sample $\{X_1, \ldots X_n\}$ with F distribution, first moment m_1 and second moment m_2, let $h(X_1, X_2) = \frac{1}{2}(X_1 - X_2)^2$ with $Eh(X_1, X_2) = \theta$. The U statistic would be

$$U = \frac{1}{\binom{n}{2}} \sum_{1 \le i_1 < i_2 \le n} \frac{1}{2}(X_1 - X_2)^2$$

and its projection would be

$$
\begin{aligned}
U - \theta &\simeq \frac{2}{n}\sum_{i=1}^{n} E((X_1 - X_2)^2/2|X_1 = x) \\
&\simeq \frac{2}{n}\sum_{i=1}^{n} E(X_1^2 - 2X_1X_2 + X_2^2)|X_1 = x)/2 - \theta \\
&\simeq \frac{2}{n}\sum_{i=1}^{n} (x^2 - 2xE(X_2) + E(X_2^2))/2 - \theta \\
&\simeq \frac{2}{n}\sum_{i=1}^{n} (x^2 - 2xm_1 + m_2)/2 - \theta
\end{aligned}
$$

and finally its asymptotic variance

$$
\begin{aligned}
A - Var(U) &= Var((X_1^2 - 2X_1m_1 + m_2)/2) \\
&= Var(X_1^2) + 4m_1^2 Var(X_1) - 2m_1 Cov(X_1^2, X_1)
\end{aligned}
$$

10.2 Variance Estimation for the U-statistic

Let's assume we want to estimate the variance of the θ using h but we aren't sure what the distribution is that the sample comes from. Just use the empirical average for sample variance! To really understand what's going on, remember that empirical average is literally just an estimation of expectation. What we do is fix the first r.v. of h as the ith sample value, then just average the non-fixed random variables over the remaining sample values.

$$
\hat{h}_1(x_1) = \underbrace{\frac{1}{n \times \cdots \times n}}_{(m-1)\text{ times}} \underbrace{\sum_{i_1=1}^{n} \cdots \sum_{i_m=1}^{n}}_{(m-1)\text{ many summations}} h(x_1, X_{i2}, \ldots, X_{im})
$$

and in general,

$$
\hat{h}_1(x_i) = \frac{1}{n^{m-1}} \sum_{i_2=1}^{n} \cdots \sum_{i_m=1}^{n} h(x_i, X_{i2}, \ldots, X_{im}).
$$

Now that we have an estimation of $h_1(x_i)$, we just need to calculate its sample variance,

$$
\hat{Var}(\hat{h}_1(x_1)) = \frac{1}{n-1}\sum_{i=1}^{n} (\hat{h}_1(x_i) - \bar{\hat{h}}_1)^2
$$

where

$$
\bar{\hat{h}}_1 = \frac{1}{n}\sum_{i=1}^{n} \hat{h}_1(x_i).
$$

10.3 U statistic example 2

Consider two independent samples $\{X_1, \ldots, X_n\}$ and $\{Y_1, \ldots, Y_m\}$ with continuous distributions F and G, respectively. Find the asymptotic distribution of the following test statistic,

$$\frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} I(X_i < Y_j),$$

under $H_0 : F = G$.

The two-sample test statistic given is

$$U = \frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} I(X_i < Y_j),$$

with kernel

$$h(X_i, Y_j) = I(X_i < Y_j),$$

and parameter

$$\theta = E_{FG}(h(X_1, Y_1)).$$

Under $H_0 : F = G$, the projection of $U - \theta$ is

$$\hat{U} = \frac{1}{n} \sum_{i=1}^{n} h_{1x}(X_i) + \frac{1}{m} \sum_{i=1}^{m} h_{1y}(Y_i),$$

where

$$h_{1x}(x) = E_{FG}(I(X_1 < Y_1)|X_1 = x) - \theta = \Pr_G(x < Y_1) - \theta = 1 - \mathbf{F}_G(x) - \theta,$$

$$h_{1y}(y) = E_{FG}(I(X_1 < Y_1)|Y_1 = y) - \theta = \Pr_F(X_1 < y) - \theta = \mathbf{F}_F(y) - \theta,$$

and $\mathbf{F}$ is the generalized CDF under the null ($H_0 : F = G$, so $\mathbf{F}_G(x) = \mathbf{F}_F(x)$ and similarly with y, hence the notation of $\mathbf{F}(x)$ will suffice). Note that $\mathbf{F}(X) \sim \text{Unif}(0, 1)$ and $\mathbf{F}(Y) \sim \text{Unif}(0, 1)$. Hence $\hat{U}$ becomes

$$\hat{U} = \frac{1}{n} \sum_{i=1}^{n} (1 - \mathbf{F}(X_i) - \theta) + \frac{1}{m} \sum_{j=1}^{m} (\mathbf{F}(Y_j) - \theta).$$

The asymptotic null distribution is,

$$U - \theta \sim AN\left(0, \frac{\xi_1^2}{n} + \frac{\xi_2^2}{m}\right),$$

where,

$$\xi_1^2 = \text{Var}(\mathbf{F}(Y_1)) = \frac{1}{12}, \qquad \xi_2^2 = \text{Var}(\mathbf{F}(X_1)) = \frac{1}{12},$$

and

$$\theta = E_{FG}(I(X_1 < Y_1)) = \frac{1}{2}.$$

Let $m + n = N$ such that,

$$N\left(\frac{1}{N}\frac{\xi_1^2}{n} + \frac{1}{N}\frac{\xi_2^2}{m}\right) = N\left(\frac{1}{12nN} + \frac{1}{12mN}\right) = \frac{N}{12}\left(\frac{m+n}{nmN}\right) = \frac{N}{12nm}\left(\frac{m+n}{N}\right) = \frac{N}{12nm},$$

and let us assume that the ratio of $\frac{n}{N}$ converges to some constant as $n \to \infty$ and $N \to \infty$, similarly for the ratio of $\frac{m}{N}$. Then it can be seen that

$$\sqrt{\frac{12nm}{N}}\left(U - \frac{1}{2}\right) \xrightarrow{d} N(0,1).$$

10.4 k-Sample U-Statistic

Let's assume now that we have a function that is defined over multiple samples,

$$h(\mathcal{X}_1; \ldots; \mathcal{X}_k) = h(X_{(1)i1}, \ldots, X_{(1)im_1}; \ldots; X_{(k)i1}, \ldots, X_{(k)im_k})$$

such that

$$E(h(\mathcal{X}_1, \ldots, \mathcal{X}_k)) = \theta.$$

The U-statistic for the function abo-*FINISH LATER*-

11 Jackknife Estimation

12 Bootstrap Estimation

13 Etc.